

Homework 7: Hypothesis testing

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Copper concentrations

(Source: Millard, S.P. and Neerchal, N.K. (2001) *Environmental Statistics with S-PLUS*, CRC Press)

Dataset `EPA.09.Ex.16.4.copper.txt` contains copper concentrations (ppb) collected at three groundwater monitoring wells: two background wells and one compliance well (USEPA, 1992c, p. 47).

- Compute summary statistics by well type (background vs. compliance).
- Use plots to compare the compliance well with the data for both background wells combined.
- Does there appear to be any evidence of contamination?
- Create a normal probability plot by combining the data from both background wells to determine whether these data may be adequately modeled with a normal distribution.
- Discuss if there is a significative difference between the two background wells.

Air quality measurements in New York

Dataset `airquality`

```
data("airquality")
```

contain daily air quality measurements in New York from May to September 1973, The data set includes the ground level ozone concentration in parts per billion from 1:00 to 3:00 PM at Roo- sevelt Island and three meteorological variables – solar radiation (`Solar.R`) in Langleys in the frequency band 4000–7700 Angstroms from 8:00 AM to 12:00 noon at Central Park, average wind speed (`Wind`) in miles per hour at 7:00 and 10:00 AM at LaGuardia Airport, and maximum daily temperature in degrees Fahrenheit at LaGuardia Airport (`Temp`).

- Compare the ground level ozone concentration in May and in August and argue if there is a significance difference.

Lake Erie

(Source: Qian, S.S. (2016) *Environmental and Ecological Statistics with R*, Chapman and Hall/CRC)

The Great Lakes Environmental Research Laboratory of the U.S. National Oceanic and Atmospheric Administration (NOAA-GLERL) routinely monitors the western basin of Lake Erie from May to October every year. The data file `LakeErie1.csv` includes all available data up to the end of 2014. Two variables are of particular interest in studying lake eutrophication. They are the total phosphorus (TP) and chlorophyll a (chl_a) concentrations. These are concentration variables and we often assume that their distributions are approximately normal.

- a. Read the data into R and use the graphics to evaluate whether TP and `chla` are normally distributed.
- b. Fit a Gaussian distribution for each variable.
- c. Use a goodness-of-fit test statistic for assessing if the data are consistent with the Gaussian distribution

In addition to NOAA-GLERL, several other institutions also have routine monitoring programs on Lake Erie. Data file `LakeErie2.csv` contains TP and chl_a concentration data collected by NOAA-GLERL, Ohio Department of Natural Resources (ODNR), and The University of Toledo (Toledo).

- d. Compare graphically the distribution of the TP concentration data from NOAA to the same from ODNR
- e. Are the two distributions different?

Uranium and total dissolved solids

(Source: Helsel, D.R. and R.M. Hirsch, R.M. (2002) *Statistical Methods in Water Resources* <http://water.usgs.gov/pubs/twri/twri4a3/>)

Consider the dataset `hh.csv`.

- a. Are uranium concentrations correlated with total dissolved solids in the following groundwater samples?
- b. If so, describe the strength of the relationship.